

Chapter 3: Physical Layer

Transmission Medium

In a data transmission system, transmission media is anything that can carry information from a source to a destination. The transmission media that are used to convey information can be classified as guided or unguided. Guided media provide a physical path along which the signals are propagated; these include twisted pair, coaxial cable, and optical fiber. Unguided media employ an antenna for transmitting through air, vacuum, or water.

Guided Transmission media

Twisted Pair Cable

A twisted pair consists of two insulated copper wires arranged in a regular spiral pattern. The purpose of twisting the wire is to eliminate electrical interference from other wires and outside sources such as motors. Twisting the wires cancels any electrical noise from the adjacent pair. The more twists per linear foot, the greater the effect. Compared to other commonly used guided transmission media (coaxial cable, optical fiber), twisted pair is limited in distance, bandwidth, and data rate.

- A straight-through cable has identical ends. A crossover cable has different ends.
- A straight through is used as a patch cord in Ethernet connections.
- A crossover is used to connect two Ethernet devices without a hub or for connecting two hubs.
- A crossover has one end with the Orange set of wires switched with the Green set.
- Odd numbered pins are always striped; even numbered pins are always solid colored.

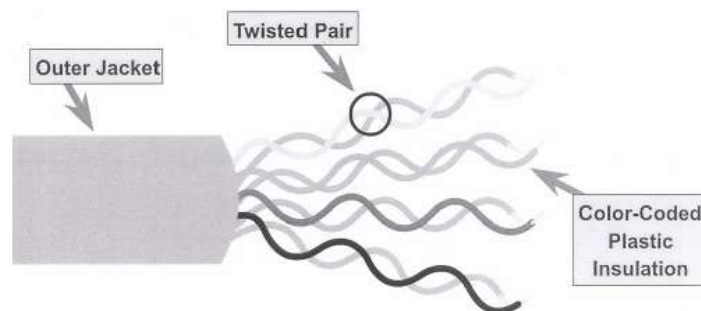


Fig: Twisted pair cable

There are two types of twisted pair cable: STP and UTP

UTP (Unshielded Twisted Pair): It is the most common medium used in telephone network, between house and local exchange (subscriber loop), within a company's buildings to private

branch exchange (PBX) and for local area networks (LAN). Because it lacks shielding UTP is not as good as STP in blocking noise and interference. The categories of UTP into 7 classes are:

- **Category 1:** Used for telephone communications. It is not suitable for transmitting data.
- **Category 2:** Capable of transmitting data at speeds up to 4 megabits per second (Mbps).
- **Category 3:** Used in 10BASE-T networks. It can transmit data at speeds up to 10 Mbps.
- **Category 4:** Used in Token Ring networks. It can transmit data at speeds up to 16 Mbps.
- **Category 5:** a tighter twist, same number of wires, just less crosstalk and higher speeds (100 Mbps)
- **Category 6 (250 MHz):** 4 pairs of 24 American Wire Gauge (AWG) copper wires. Category 6 cable is currently the fastest standard for UTP.
- **Category 7 (600 MHz)** is upcoming.

STP (Shielded Twisted Pair): It is covered with a foil shield to reduce cross talk and interference. It provides better performance but more expensive and difficult to work than UTP. STP can handle high-speed transmission and used in token ring network.

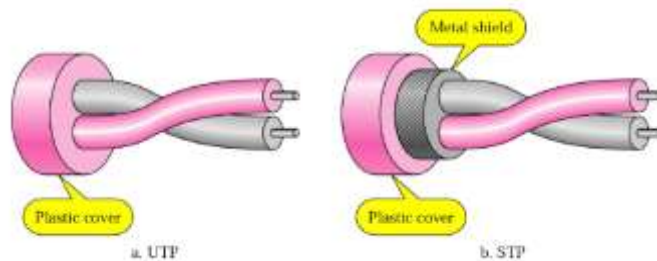


Figure: UTP and STP

Advantages

- Protect against cross talk & interference
- Easy to add computers to network
- Well understood technology
- Less expensive

Disadvantages

- Susceptibility to noise
- Least secure
- Distance limitations
- Requires more expensive hubs

Coaxial Cable

Coaxial cable, like twisted pair, consists of two conductors, but is constructed differently to permit it to operate over a wider range of frequencies. It consists of a hollow outer cylindrical conductor that surrounds a single inner wire conductor (Figure). The inner conductor is held in place by either regularly spaced insulating rings or a solid dielectric material. The outer conductor is covered with a jacket or shield. A single coaxial cable has a diameter of from 1 to 2.5 cm. Coaxial cable can be used over longer distances and support more stations on a shared line than twisted pair.

Applications

- Television distribution
- Long-distance telephone transmission
- Short-run computer system links
- Local area networks

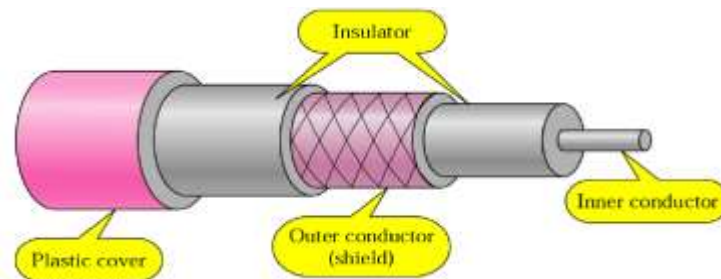


Figure: Coaxial cable

Advantages

- Transmits up to 10Mbps over 500m
- Easy to install
- Low maintenance
- Good resistance to noise over long distances

Disadvantages

- Inflexible
- Low security
- Limited distance

Optical Fiber

An optical fiber cable has a cylindrical shape and consists of three concentric sections: the core, the cladding, and the jacket (Figure). The **core** is the innermost section and consists of one or more very thin strands, or fibers, made of glass or plastic. Its own cladding, a glass or plastic coating that has optical properties different from those of the core surrounds each fiber. The interface between the core and cladding acts as a reflector to confine light that would otherwise escape the core. The outermost layer, surrounding one or a bundle of cladded fibers, is the **jacket**. The jacket is composed of plastic and other material layered to protect against moisture, abrasion, crushing, and other environmental dangers.

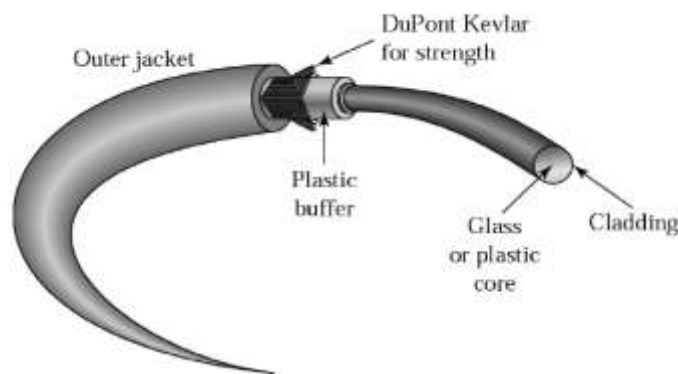


Figure: Optical Fiber

We must have a light source in order to send the data. It sends a ray of light for each 1 bit, and no light for a 0 bit. We must have a detector to detect the light signal. The detector emits an electric pulse for each light ray it detects. The slowest part of the system is the conversion that happens at either end.



Figure: Transmission

Light propagation modes

Step-index multimode: It's refraction index is constant for the fibre core, distance to core's centre does not matter that implies different path lengths for light rays, making reception difficult. It present a thicker core (hundreds of μm) so cheaper fiber.

Graded-index multimode: It's refraction index decreases from the core centre to edges that offer a better focusing of the rays, so a lower attenuation and easier reception.

Single mode (mono-mode): It has direct path for light ray, no loss, no attenuation, but more expensive

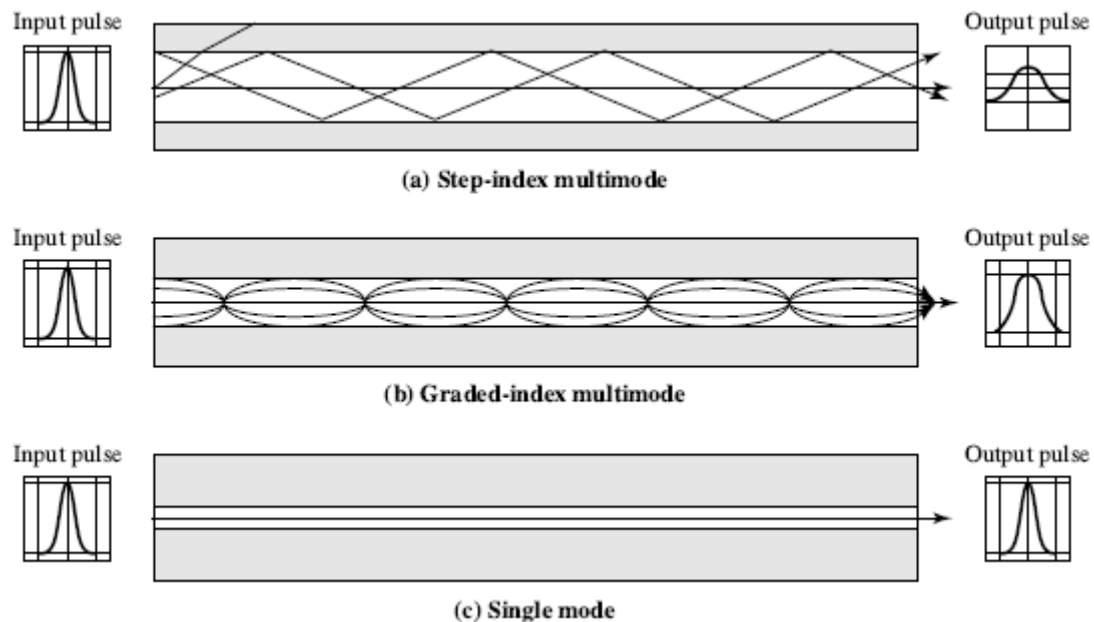


Figure: Optical fiber transmission mode

Advantages

- High data rate and wide bandwidth
- Low attenuation (data loss)
- Longer distance –up to 5 km with Multimode fiber or over 25 km with Single Mode fiber
- Small cable diameter fits anywhere
- No sparks if cut and no shock hazard
- Secure communications
- Longer life expectancy than copper or coaxial cable
- Cabling of the future

Disadvantages

- Expensive
- Difficult to install
- Require two cables to transmit & receive data
- Require special connections

Unguided Transmission Media

Bluetooth

- Developed by SIG (Special Interest Group)
- The IEEE 802.15 Standard defines Bluetooth.
- It defines Wireless PAN operable in an area of room or a hall.
- Used Technology Called FHSS (Frequency Hopping Spread Spectrum)
- When two Bluetooth devices notice each other they create a network called a Piconet

Bluetooth Architecture

- Piconet
- Scatternet

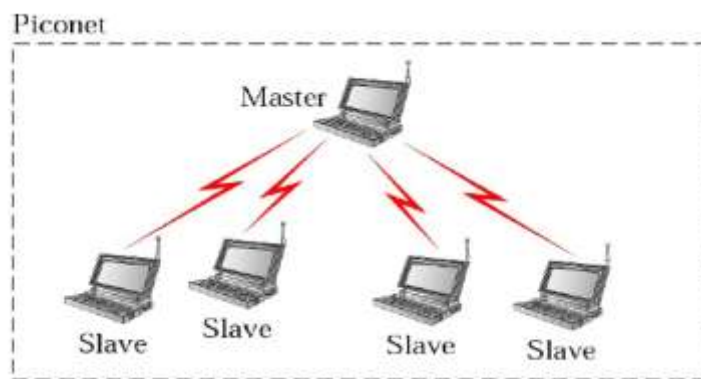


Figure: Piconet

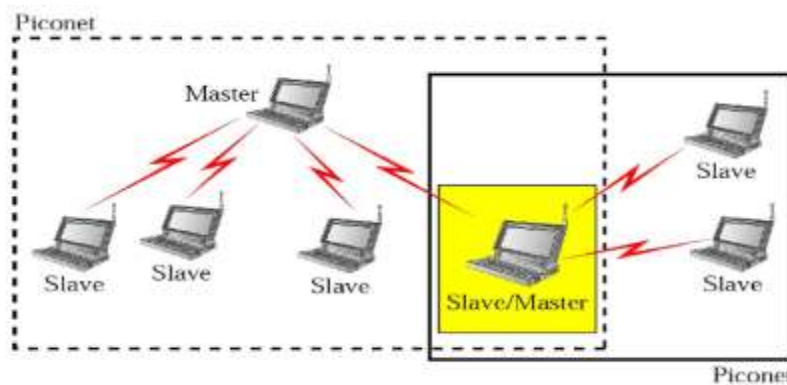


Figure: Scatternet

Radio wave

Radio waves are a type of electromagnetic radiation with wavelengths in the electromagnetic spectrum longer than infrared light.

The principal difference between broadcast radio and microwave is that the former is omnidirectional and the latter is directional. Thus broadcast radio does not require dish-shaped antennas, and the antennas need not be rigidly mounted to a precise alignment.

Applications

- TV & radio broadcasting
- Cordless phones

Infrared

Infrared communications is achieved using transmitters/receivers (transceivers) that modulate non-coherent infrared light. Transceivers must be within the line of sight of each other either directly or via reflection from a light-colored surface such as the ceiling of a room. One important difference between infrared and microwave transmission is that the former does not penetrate walls. Thus the security and interference problems encountered in microwave systems are not present. Furthermore, there is no frequency allocation issue with infrared, because no licensing is required. Unguided infrared waves are widely used for short-range communication. The remote controls used for televisions, VCRs, and stereos all use infrared communication. They are relatively directional, cheap, and easy to build. They have a major drawback that they do not pass through solid objects.

Microwave

Microwave links are widely used to provide communication links when it is impractical or too expensive to install physical transmission media. Two properties of microwave transmission restrict its use. First, microwave travels in straight path and will not follow earth's curvature. Second atmospheric condition and solid objects interfere with microwave. For example, they cannot travel through buildings.

Applications

- Long-distance telephone communication
- Mobile phones
- Television broadcast

Satellite Communication

Satellite transmission is a microwave transmission in which one of the stations is the satellite orbiting the earth. A microwave beam is transmitted to the satellite from the ground. The two stations can use a satellite as a relay station for their communication. Receiver and transmitter in satellite is known as transponder. One Earth Station sends a transmission to the satellite. This is called a Uplink. The satellite Transponder converts the signal and sends it down to the second earth station. This is called a Downlink.

The advantages of satellite communication over terrestrial communication are:

- The coverage area of a satellite greatly exceeds that of a terrestrial system.
- Transmission cost of a satellite is independent of the distance from the center of the coverage area.
- Satellite-to-Satellite communication is very precise.
- Higher Bandwidths are available for use.

The disadvantages of satellite communication are:

- Launching satellites into orbit is costly.
- Satellite bandwidth is gradually being used up.
- There is a larger propagation delay in satellite communication than in terrestrial communication.

Wireless Propagation

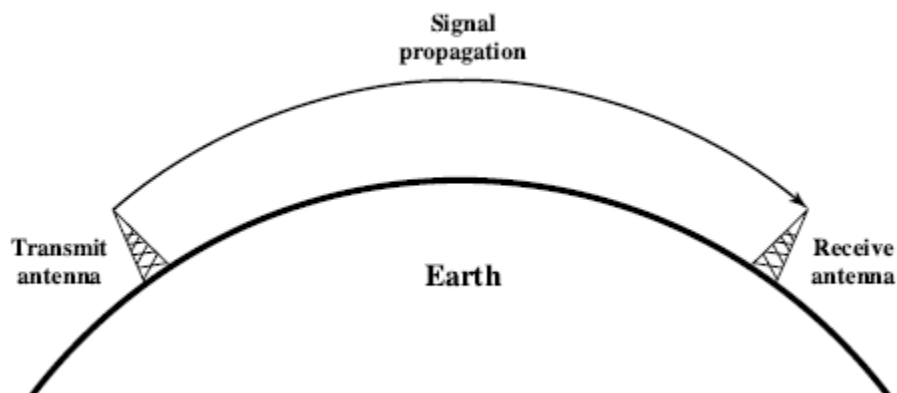


Figure: Ground wave propagation

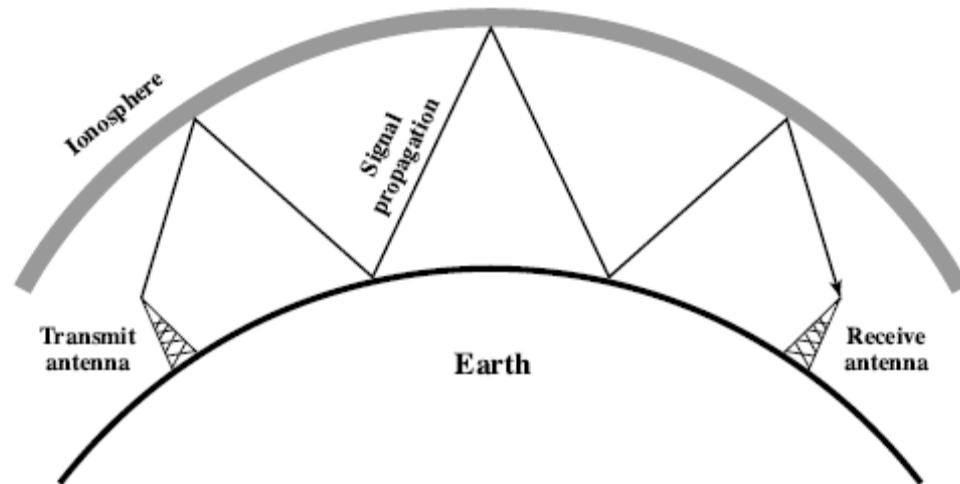


Figure: Sky wave propagation

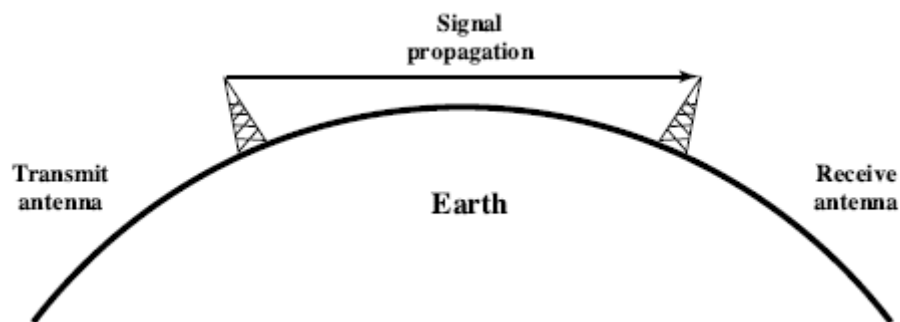


Figure: Line of Sight Propagation

Switching

Every time in computer network you access the internet or another computer network outside your immediate location, your messages are sent through a maze of transmission media and connection devices. The mechanism for moving information between different computer network and network segment is called switching in computer network.

There are three different types of switching:

- Message switching
- Circuit switching
- Packet switching

Message switching

It does not need a physical path to set up between the sender and receiver. The whole message (or block of data) is sent to the switching office. Once it has been received, it is inspected for errors and is then sent to the next switching office. This method is not used anymore.

Circuit switching

Conceptually, when you or your computer places a telephone call, the switching equipment within the telephone system seeks out a physical path all the way from your telephone to the receiver's telephone. This technique is called **circuit switching**. This used to be done by a person at a switchboard. Now it is done automatically. Setting up the circuit can still take time, depending on how far the call is going and how many switches it passes through.

An important property of circuit switching is the need to set up an end-to-end path *before* any data can be sent. The elapsed time between the end of dialing and the start of ringing can easily be 10 sec, more on long-distance or international calls. During this time interval, the telephone system is hunting for a path, as shown in Fig. 2.16(a). Note that before data transmission can even begin, the call request signal must propagate all the way to the destination and be acknowledged.

Packet switching

With this technology, packets are sent as soon as they are available. There is no need to set up a dedicated path in advance. It is up to routers to use store-and-forward transmission to send each packet on its way to the destination on its own. This procedure is unlike circuit switching, in which the result of the connection setup is the reservation of bandwidth all the way from the sender to the receiver. All data on the circuit follows this path. Among other properties, having all the data follow the same path means that it cannot arrive out of order. With packet switching there is no fixed path, so different packets can follow different paths, depending on network conditions at the time they are sent, and they may arrive out of order.

Two Types of Packet Switching

- Datagram Packet Switching
- Virtual Circuit Packet Switching

Packet Switching: Datagram Packet Switching

- No need to establish the connection between the source and destination
- Route chosen on packet by packet basis.
- Packets may be stored until delivered => (Store and Forward)
- Different packets may follow different routes.
- Packets may arrive out of order at the destination.

Packet Switching: Virtual Circuit Switching

- Route is chosen at the start of session and it is only a logical connection.
- All Packets associated with a session follow the same path.
- Packets are labeled with a virtual circuit identifier designated the route.
- The VC number must be unique on a given link.
- Packets are forwarded more quickly. (No Routing Decisions)
- Example : Asynchronous Transfer Mode

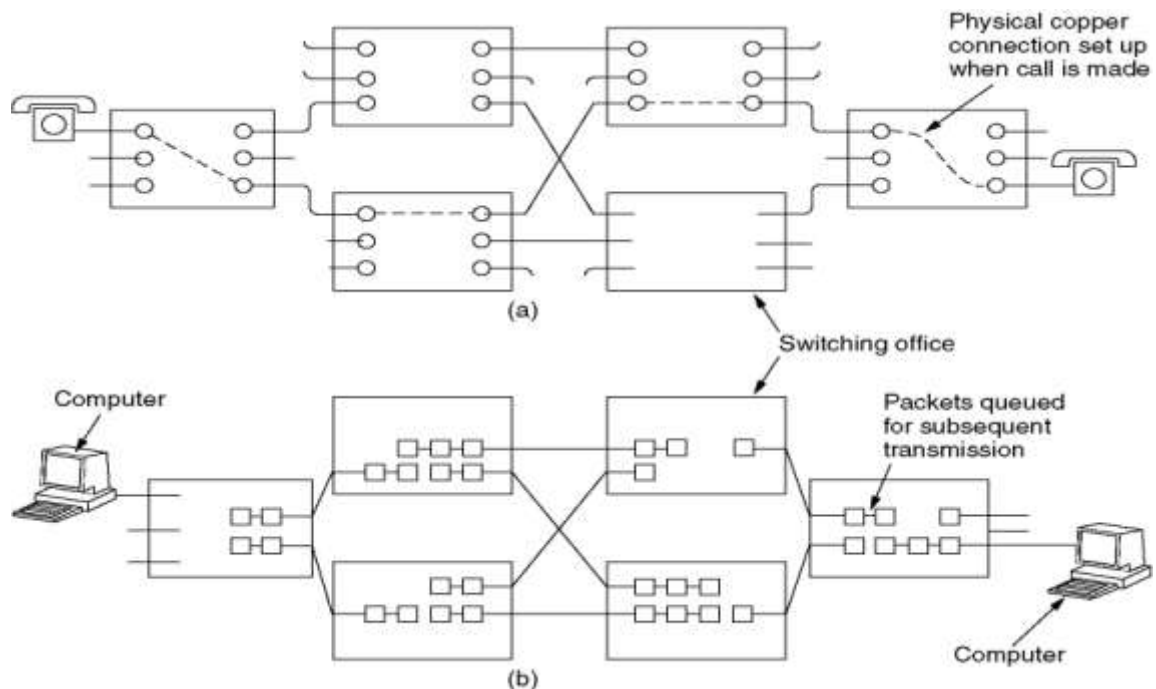


Figure: a) circuit switching b) packet switching

Comparison of Circuit and Packet switching

Item	Circuit-switched	Packet-switched
Call setup	Required	Not needed
Dedicated physical path	Yes	No
Each packet follows the same route	Yes	No
Packets arrive in order	Yes	No
Is a switch crash fatal	Yes	No
Bandwidth available	Fixed	Dynamic
When can congestion occur	At setup time	On every packet
Potentially wasted bandwidth	Yes	No
Store-and-forward transmission	No	Yes
Transparency	Yes	No
Charging	Per minute	Per packet

ISDN

The evolution to the existing telecommunications networks and specialized carrier facilities to integrated digital networks is based on two technological developments:

- Digital switching
- Digital transmission

Both technologies are well established. But IDN provided a revolutionary idea of integrating the functions of these two. As the use of distributed processing and data communications has grown, the evolution of an all-digital network has been pulled by the need to provide a framework for ISDN (Integrated Service Digital Network). It is ITU Standard For global Digital Communication. It allows the complete integration of both Voice, Video and Data within a single system.

Principles of ISDN

- Support of voice and non-voice applications using a limited set of standardized facilities
- Support for switched and non-switched applications
 - both circuit-switched and packet-switched connections
 - also supports non-switched services in the form of dedicated lines

- Reliance on 64-kbps connections
 - fundamental block of ISDN
 - chosen because it was the standard rate for digitized voice
- Intelligence in the network
 - sophisticated network management and maintenance capabilities
- Layered protocol architecture
 - Already developed standards for OSI may be used for ISDN (e.g. X.25)
 - New ISDN standard can be based on existing ones (LAPD based on LAPB)
- Variety of configurations
 - More than one physical configuration is possible for implementing ISDN

ISDN Channels

B-channel

- 64 kbps
- basic user channel
- can carry digital data, PCM-encoded digital voice, or a mixture of lower-rate traffic
- supports circuit-switched and packet-switched

D-channel

- 16 or 64 kbps
- carries signaling information to control circuit switched calls on B-channel such as who is calling, type of call and calling what number

H-channel

- 384 (H0), 1536 (H11), 1920 (H12) kbps
- is a high-speed channel
- can be used as a single trunk or subdivided by the user fast fax, video, high-speed data, high-quality audio and multiplexed information streams at lower data rates

ISDN Services Types

Also called ISDN Interface

- Basic Rate Interface(BRI)
 - Connection from ISDN office to the user location provides for access to 3 channels.
 - 2B channels + 1 D channel
 - $2 \times 64 + 16 = 144$ kbps (192 kbps total)



- Primary Rate Interface(PRI)
 - Especially for LAN
 - T1 connection: 23B+D
 - American Standards
 - Data rate $23 \times 64\text{Kbps} + 64\text{Kbs} + 8\text{bits header}$
 - Information = $1544\text{kbs} = 1.544\text{Mbps}$
 - E1 Connection: 30B+D
 - European Standards
 - $30 \times 64\text{Kbps} + 64\text{Kbps} + 64\text{Kbps}$
 - $2048\text{Kbps} = 2.048\text{Mbps}$
 - Last D channel for Framing and Synchronization in E1 Connections

ISDN Architecture

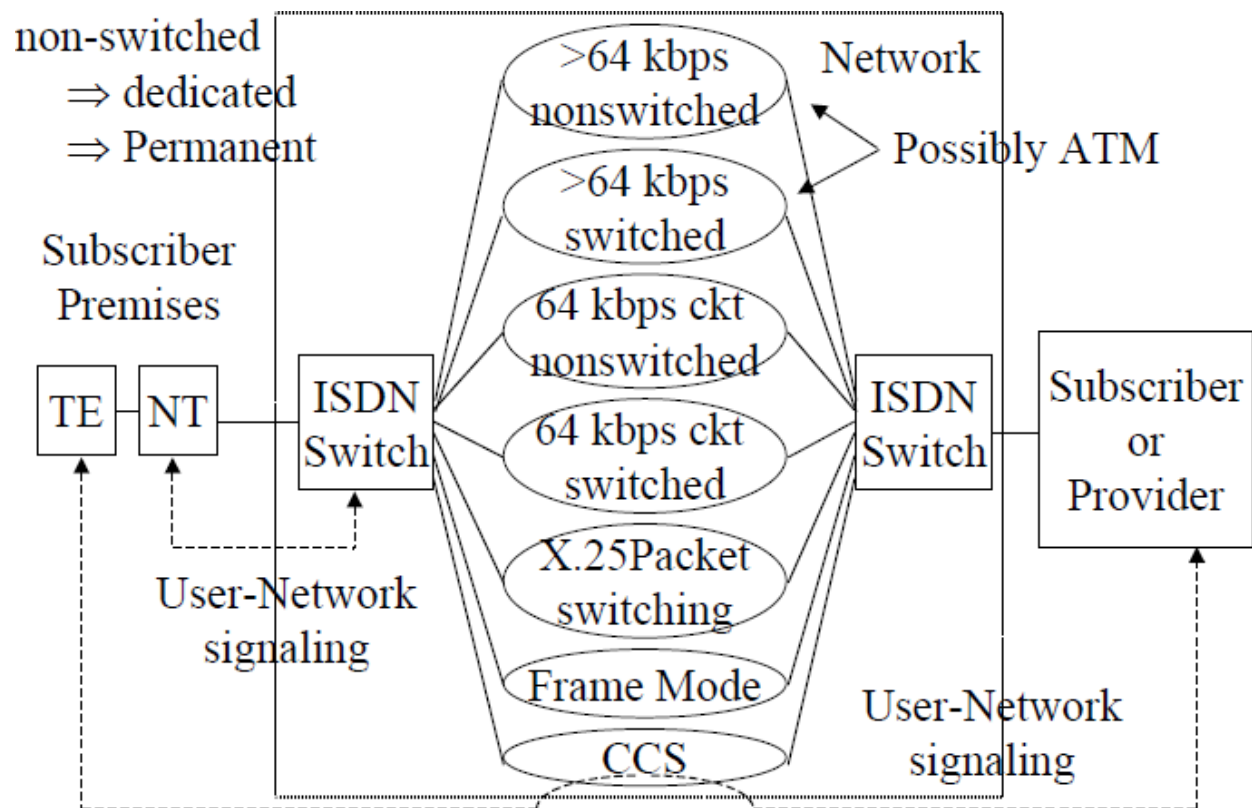


Figure: ISDN Architecture

ISDN supports a new physical connector for users, a digital subscriber line, and a variety of transmission services. Physical interface provides a standardized means of attaching to the

network. The interface supports a *basic* service consisting of three times multiplexed channels, two at 64 kbps and one at 16 kbps. In addition, there is a *primary* service that provides multiple 64- kbps channels. For both basic and primary service, an interface is defined between the customer's equipment (TE) and a device on the customer's premises, known as a network termination (NT). The subscriber line is the physical path from the subscriber's NT to the ISDN central office. ISDN central office connects subscriber lines to the digital network, providing access to lower-layer transmission facilities.

- Circuit-switched capabilities
 - Same facility provided by other digital-switched telecom. net's (64 kbps)
- Non-switched capabilities
 - 64 kbps dedicated link, higher rates in B-ISDN using PVC in ATM tx
- Switched capabilities
 - high speed (>64 kbps) switched connections using ATM in B-ISDN
- Packet-switched capabilities
 - resembles packet-switched service provided by other data networks
- Frame-mode capabilities
 - a service that supports frame relay
- Common-channel signaling capabilities
 - used to control the network and provide call management

ISDN Signaling

The signaling method will be Common Channel Signaling (CCS).

- The signaling information relating to a multiplicity of channels or functions or for network management is conveyed over a single channel by addressed messages.
- CCS will be able to operate in two modes:
 - **associated mode** (i.e. signaling points, that are the origin and destination points of the messages are directly interconnected by a link)
 - **quasi-associated mode** (i.e. messages pass through one or more signaling points other than those which are the origin or destination of the messages. The path is pre-determined).

Bandwidth

Bandwidth is the maximum amount of data that can be carried from one point to another in a given time period (usually in second). For Digital Devices, it refers to channel capacity. Bandwidth is expressed in bits per second (bps).

For Analog Devices, it refers to the range of frequencies in a composite signal or the range of frequencies that a channel can pass. Bandwidth is expressed in cycle per sec OR Hz.

Throughput

Throughput is the rate of successful message delivery over a communication channel. It is usually measured in bits per second.

Example:

A network with bandwidth of 10 Mbps can pass only an average of 12,000 frames per minute with each frame carrying an average of 10,000 bits. What is the throughput of this network?

Solution,

We can calculate the throughput as

$$\text{Throughput} = \frac{12,000 \times 10,000}{60} = 2 \text{ Mbps}$$

The throughput is almost one-fifth of the bandwidth in this case.

Bandwidth versus throughput

Bandwidth is the theoretical capability of the connection; the throughput is the actual data rate of a specific application.

Delay

Delay is the time required for a signal to traverse the network.

Propagation Delay

Propagation speed: speed at which a bit travels through the medium from source to destination

Propagation delay: Time taken for the first bit to travel from the sender to the receiver

$$\text{Propagation delay} = \text{Distance (Link Length)} / \text{Propagation speed}$$

Transmission Delay

Transmission speed: The amount of time from the beginning until the end of a message transmission

Transmission delay: Amount of time required to push all of the packet's bits into the wire

$$\text{Transmission Delay} = \text{Message size} / \text{bandwidth}$$

It is also called Store and Forward Delay. It is function of Packet Length, nothing to do with distance between the nodes.

Example

What are the propagation time and the transmission time for a 2.5-kbyte message (an e-mail) if the bandwidth of the network is 1 Gbps? Assume that the distance between the sender and the receiver is 12,000 km and that light travels at 2.4×10^8 m/s.

Solution

We can calculate the propagation and transmission time as

$$\text{Propagation time} = \text{Distance} / \text{Speed}$$

$$\text{Transmission time} = \text{Packet Size} / \text{bandwidth of link}$$

$$\begin{aligned}\text{Propagation time} &= \frac{12,000 \times 1000}{2.4 \times 10^8} = 50 \text{ ms} \\ \text{Transmission time} &= \frac{2500 \times 8}{10^9} = 0.020 \text{ ms}\end{aligned}$$

Latency

Network latency is the delay that is introduced by the network

$$\text{Latency} = \text{Propagation delay} + \text{Transmission delay} + \text{Queuing time} + \text{Processing time}$$

Low Latency: Network that experiences Low Delay

High Latency: Network that experience High Delay

Jitter

Jitter is the variation in the time between packets arriving, caused by network congestion, timing drift, or route changes. The causes of jitter are electromagnetic interference (EMI) and crosstalk with other signals.

Bandwidth Delay Product

Product of a data link's capacity (in bits per second) and its round-trip delay time (in seconds).

A network with a large bandwidth-delay product $> 10^5$ bits is commonly known as a long fat network.

Baseband and Broadband

Baseband: digital signals sent through direct current (DC) pulses applied to a wire. It requires exclusive use of wire's capacity. Baseband systems can transmit one signal at a time.

Broadband: signals modulated as radiofrequency (RF) analog waves that use different frequency ranges. It does not encode information as digital pulses.